



Real Estate Price Prediction via GBDT, Coordinate encoding, and Stacking

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Dataset & Motivation

USA Housing Price Dataset

USA House Prices

Unlocking the Secrets of Real Estate: Analyzing House Prices with Data



Dataset & Motivation

USA Housing Price Dataset

Characteristics)

- Mixture of Numeric Data & String Data

Upper 0.10% threshold (q=0.999): \$4,393,229
Outliers above threshold: 5

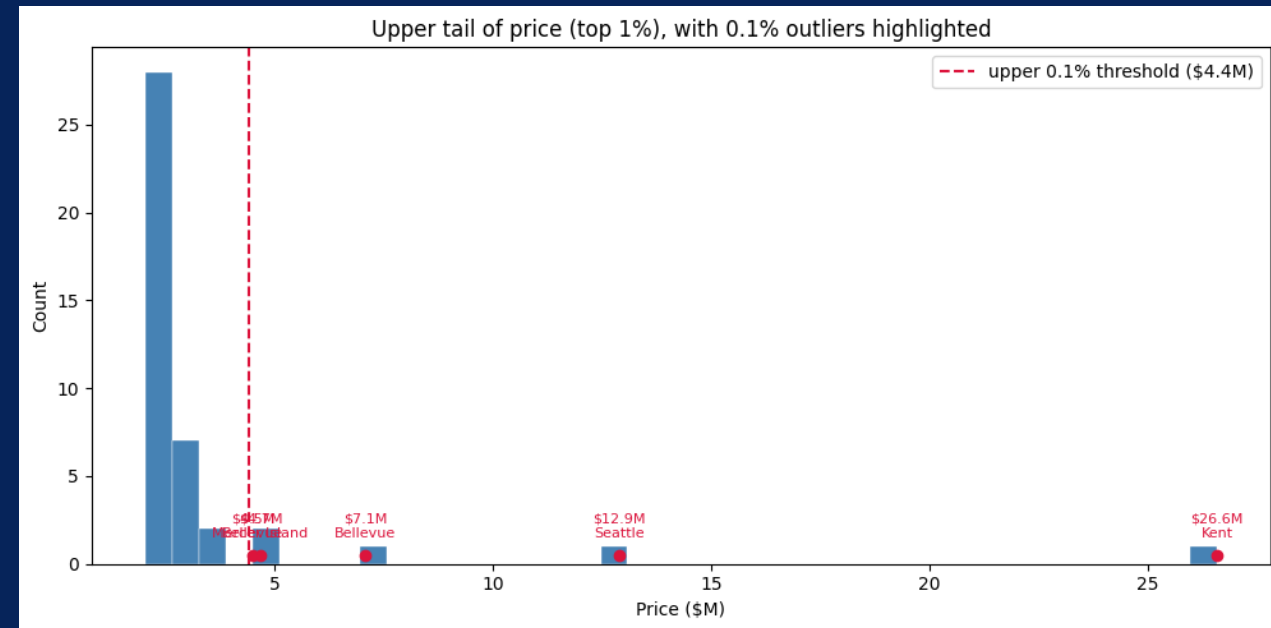
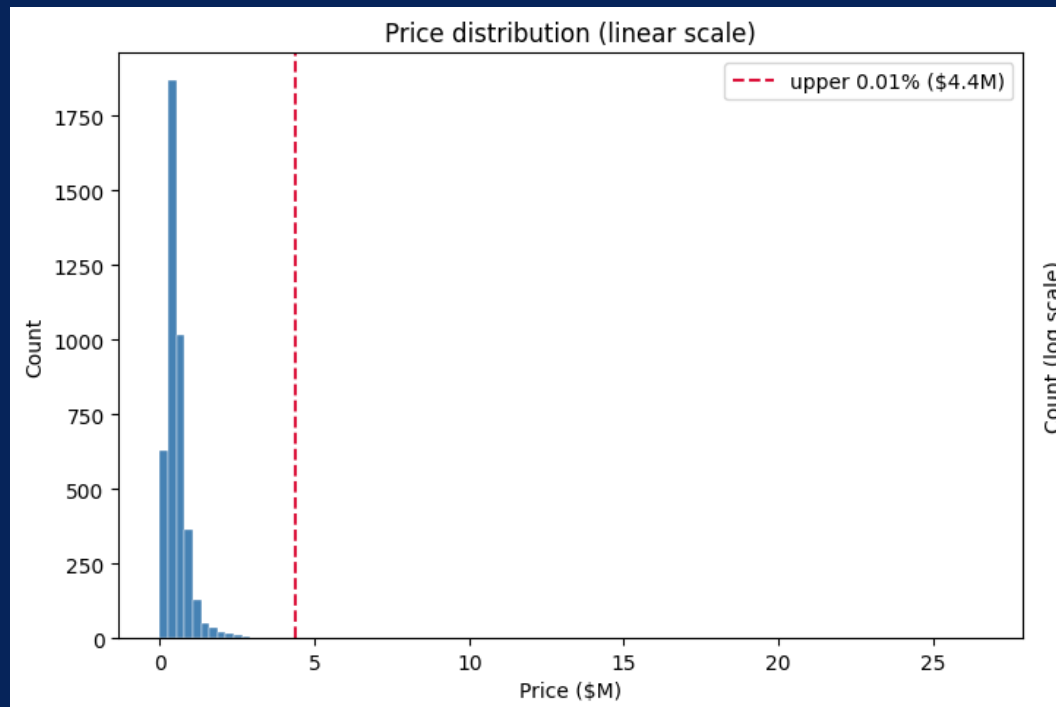
	price	bedrooms	bathrooms	sqft_living	city	statezip
3890	26590000.0	3.0	2.00	1180	Kent	WA 98031
3886	12899000.0	3.0	2.50	2190	Seattle	WA 98199
1826	7062500.0	5.0	4.50	10040	Bellevue	WA 98004
2194	4668000.0	5.0	6.75	9640	Mercer Island	WA 98040
2301	4489000.0	4.0	3.00	6430	Bellevue	WA 98004

Dataset & Motivation

USA Housing Price Dataset

Characteristics)

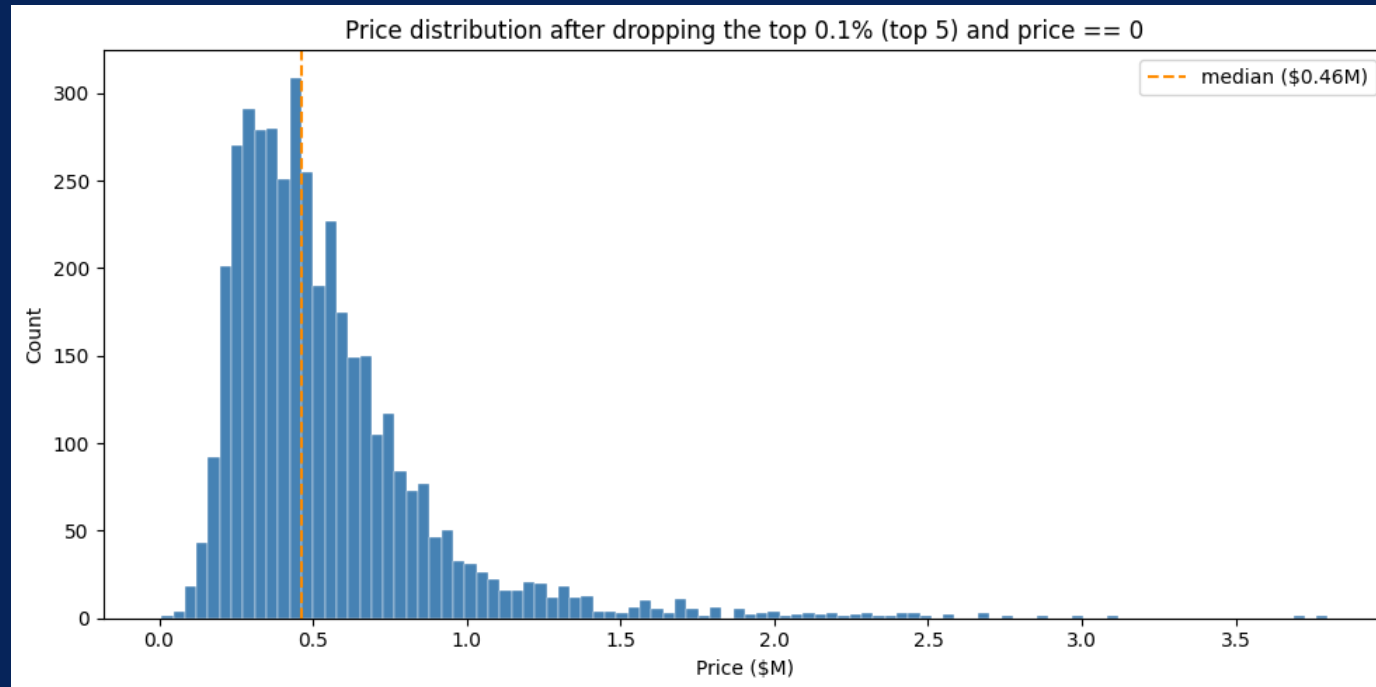
- High Skewness



Dataset & Motivation

USA Housing Price Dataset

- We decided to drop only top-5 outliers! (4091 -> 4086 rows)
- Resemble real-life data noises.

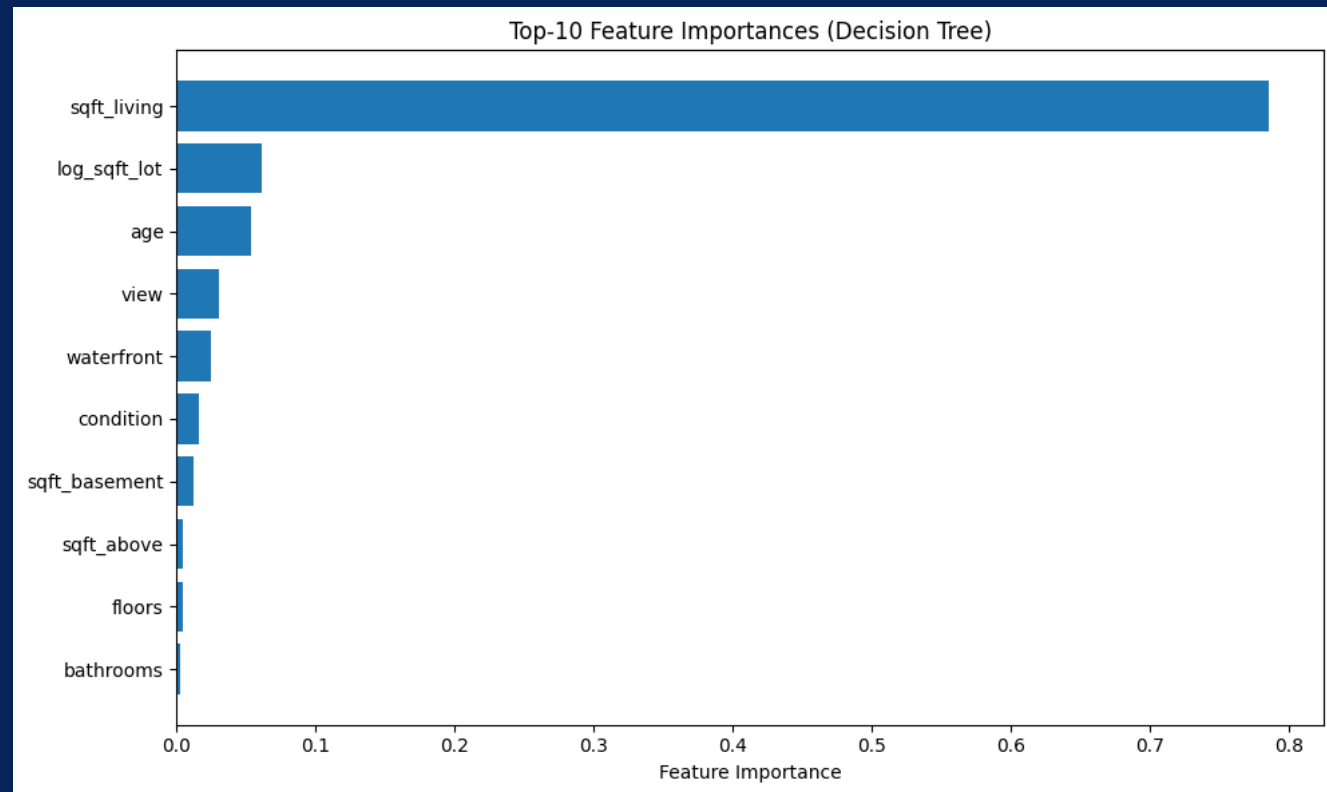


Dataset & Motivation

USA Housing Price Dataset

Characteristics)

- High Feature Imbalance

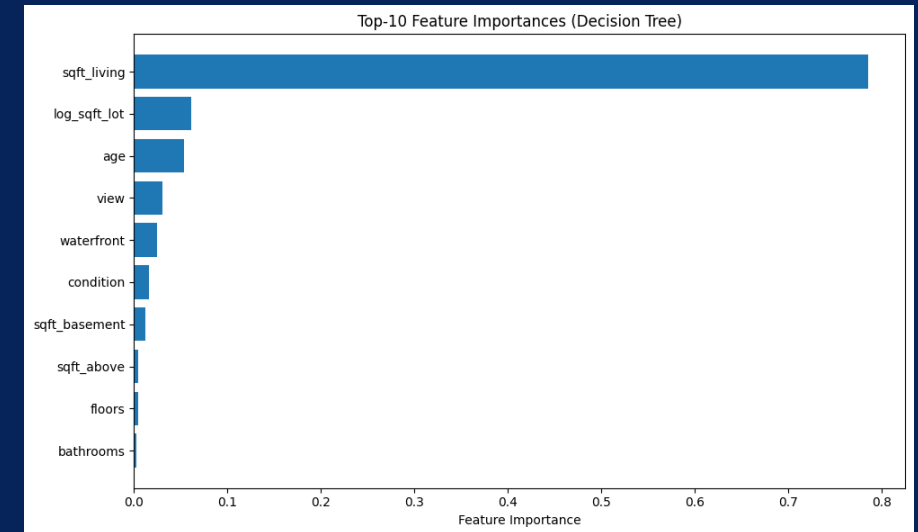


Dataset & Motivation

Theory: Model only focus on dominant feature!

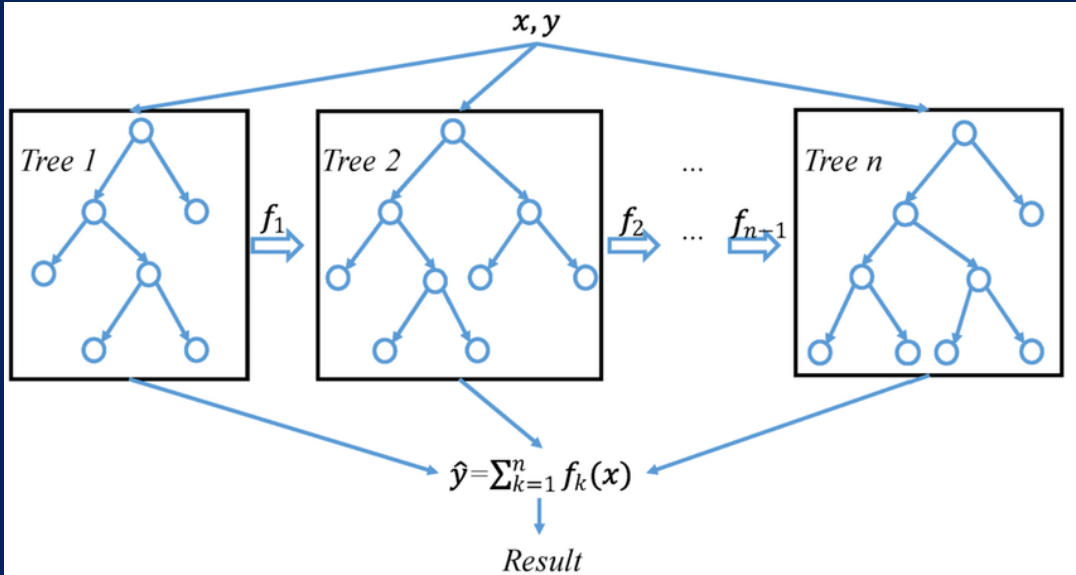
Idea: More sophisticated structure might capture deeper relationship on other features.

-> Model candidate: MLP, Decision Tree, Random Forest, mGBDT

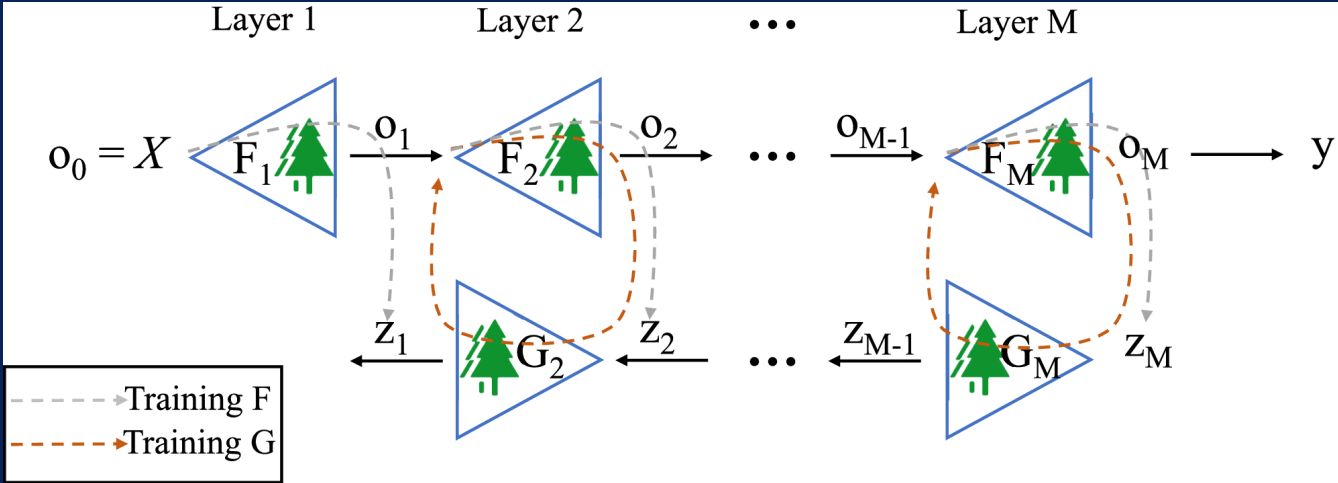


Background

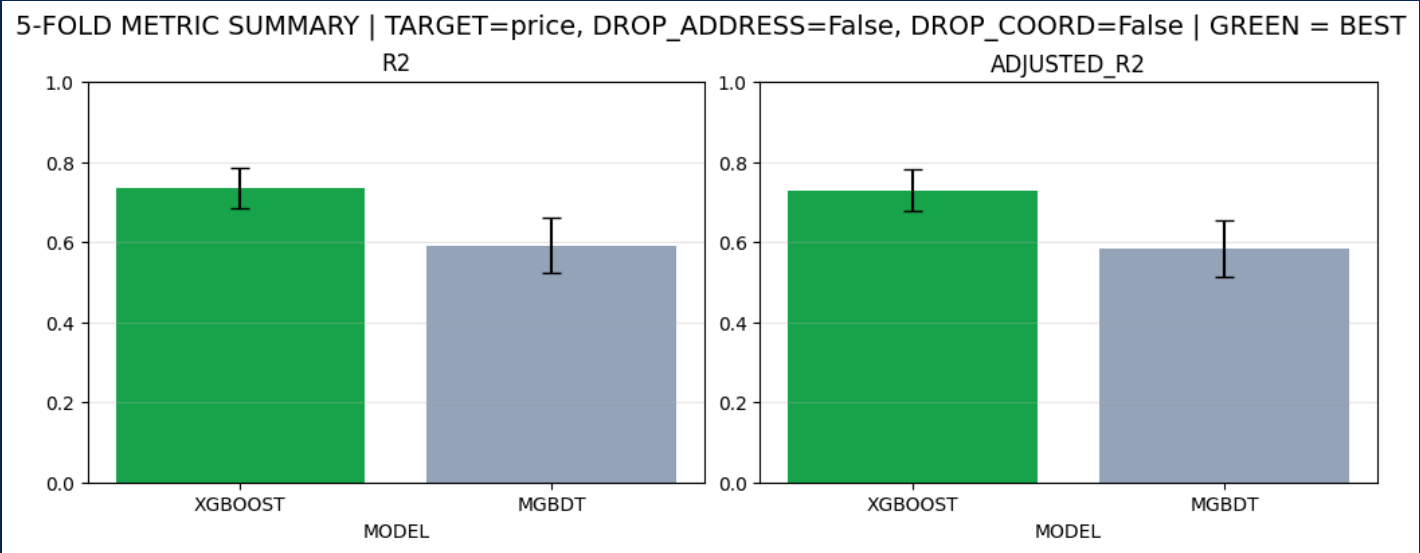
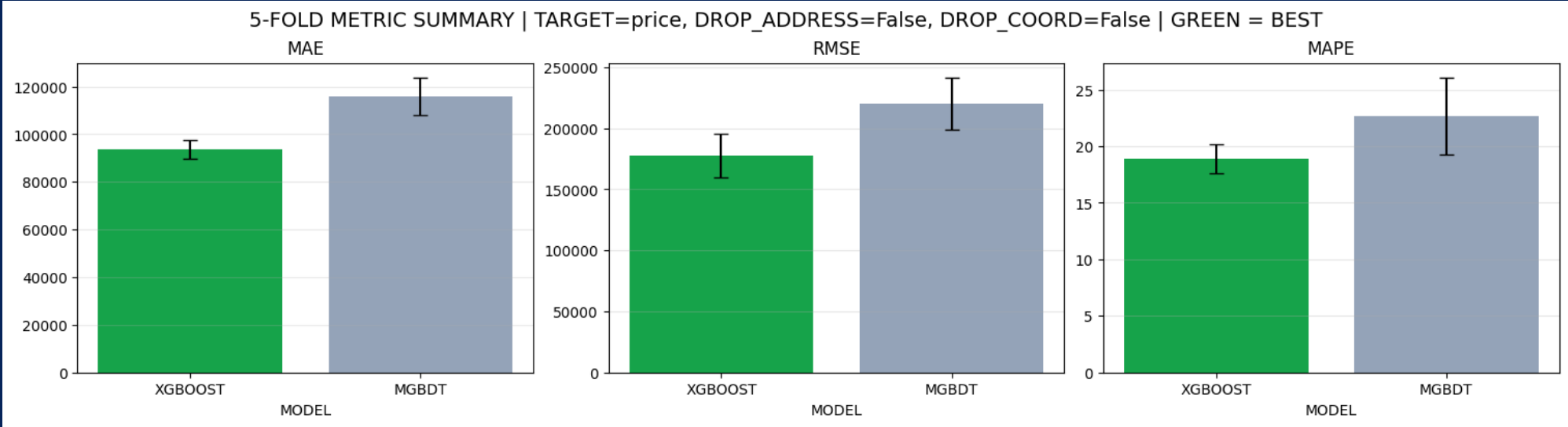
GBDT (XGBoost)



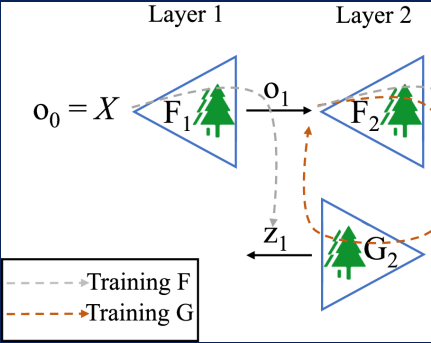
mGBDT



Background

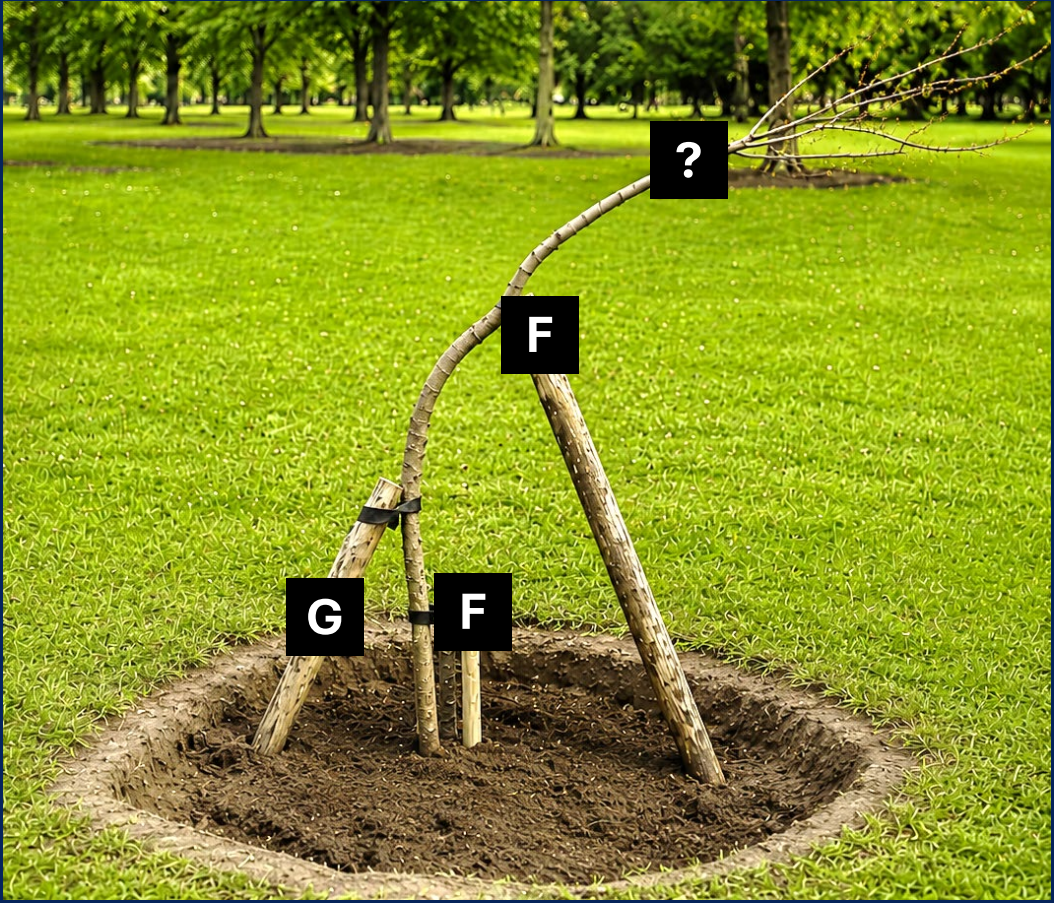
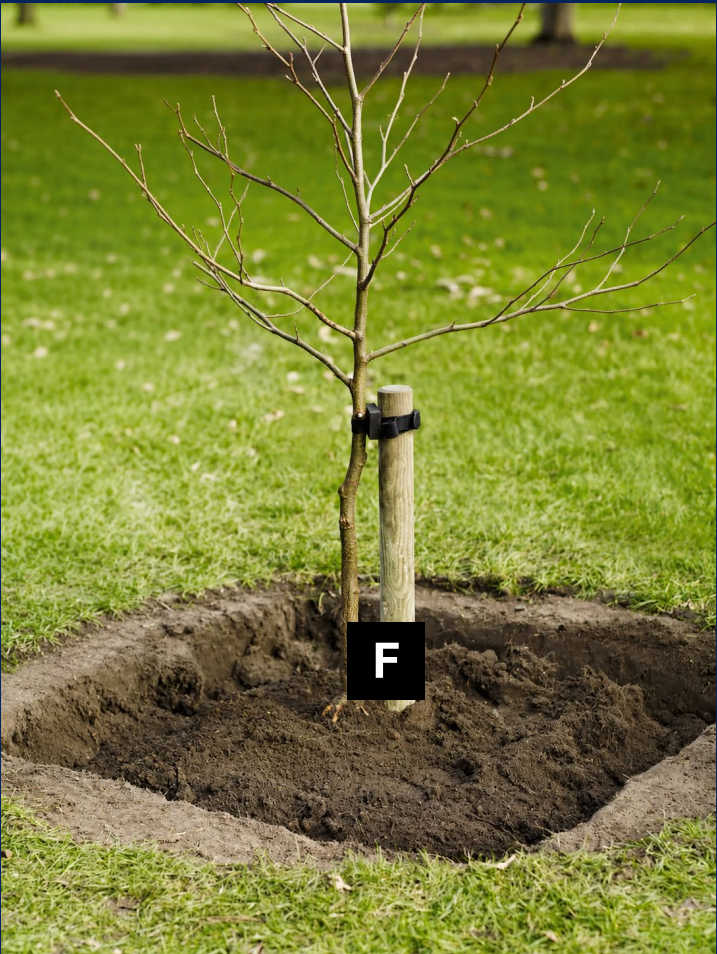
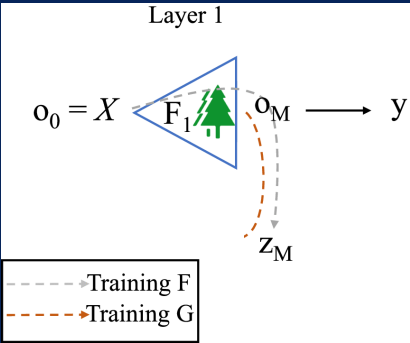


Background



GBDT (XGBoost)

mGBDT



Experiment

Baseline: Random Forest with Categorical Encoding

Categorical Encoding

- Convert literal data into one-hot vector/integer numeric

Ex) ["Seattle", "Seattle", "Ulsan", "Kent"] -> [1, 1, 2, 3]

Alternative: Target Encoding

- Convert literal data into statistics of that category

Ex) ["Seattle", "Seattle", "Ulsan", "Kent"] -> [(average house price of Seattle), ...]

Experiment

Proposal 1: Coordinate encoding

Observation: Dataset has detailed “address” per housing

Ex) “City”, “Street”, “Statezip” (=code for region)

Idea: Addresses can be converted into coordinates (latitude & longitude)!

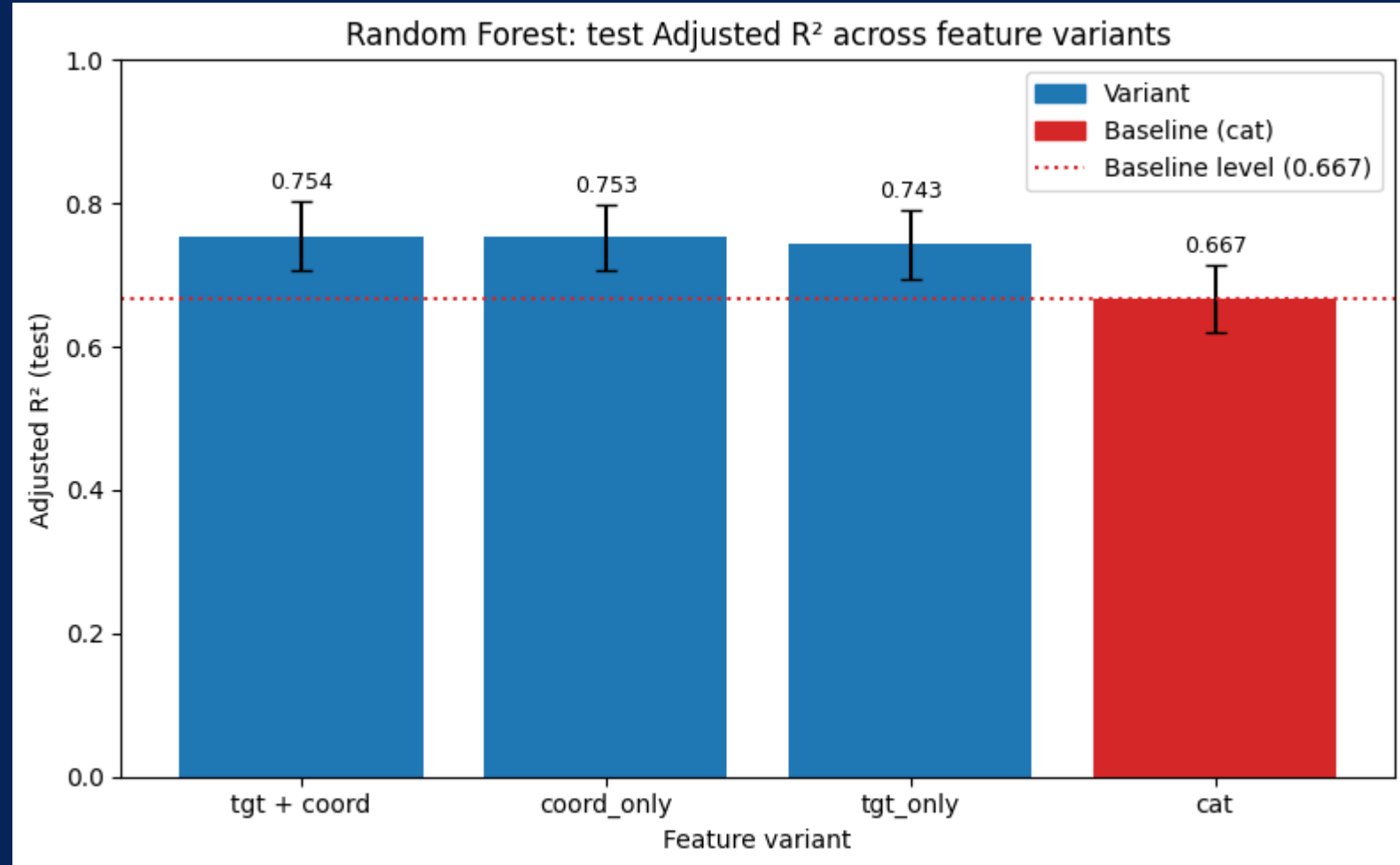
Ex) 9245-9249 Fremont Ave N, Seattle, WA 98103,USA -> (47.696..., -122.350...)



(We utilized Geocode API for mass convert)

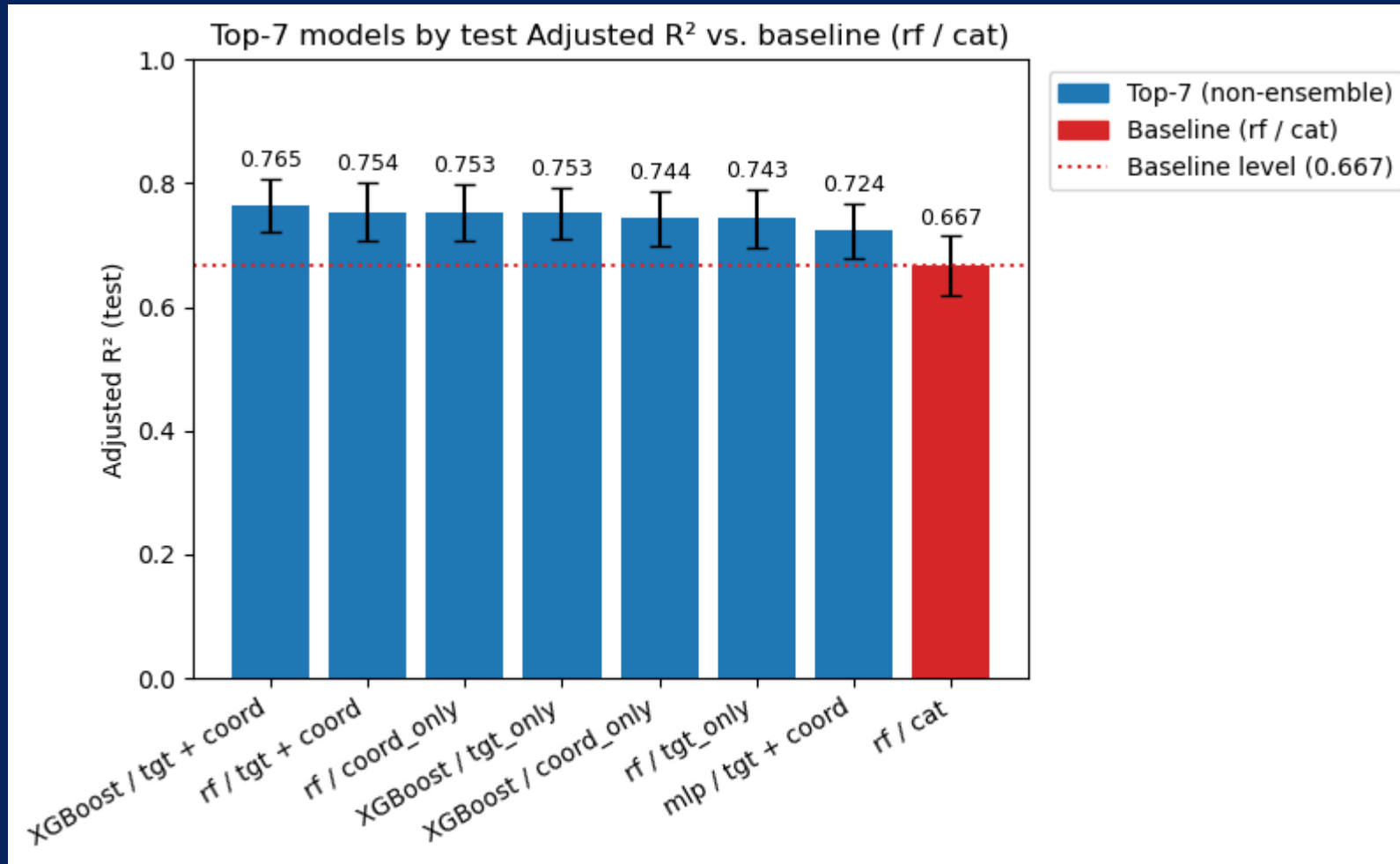
Result

Result 1: Both target & coordinate encoding works well



Result

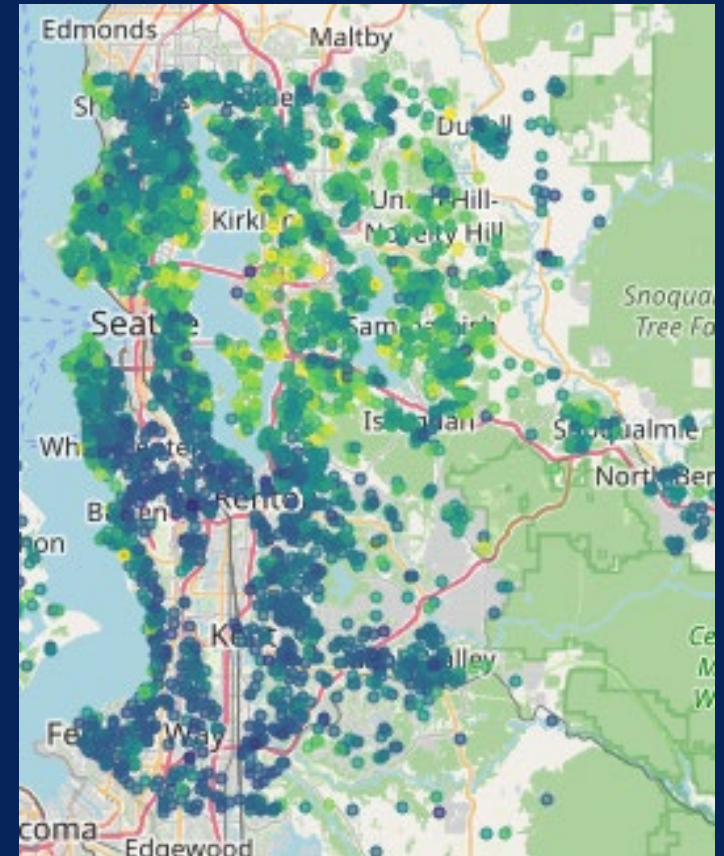
Result 1: Performance gain from coordinate encoding varies



Result

Result 1: Performance gain from coordinate encoding varies

- Both target encoding & coordinate encoding outperforms categorical encoding
- Difference between target encoding vs coordinate encoding was minimal
- Possibly due to the nature of house distribution
 - Infrastructure, oceans, mountains, etc.



Experiment

Proposal 2: Price-per-sqft

Observation:

We can regularize some outliers by dividing it with sqft_living

price	sqft_living	price_per_sqft
\$2,888,000	8670	\$333
\$2,750,000	4430	\$621
\$2,700,000	5305	\$509
\$2,700,000	4420	\$611
\$2,680,000	4290	\$625
\$2,560,498	1710	\$1,497
\$2,555,000	5300	\$482
\$2,475,000	4340	\$570
\$2,466,350	6390	\$386
\$2,458,000	6500	\$378
\$2,453,500	4730	\$519
\$2,400,000	4140	\$580
\$2,400,000	3410	\$704
\$2,400,000	5480	\$438
\$2,367,000	3530	\$671
\$2,351,956	5010	\$469

Experiment

Proposal 2: Price-per-sqft

Observation:

We can regularize some outliers by dividing it with sqft_living

-> “Price” outliers might not be “Price-per-sqft” outliers!

price_rank	per_sqft_rank	rank_change
10	860	-850
11	38	-27
12	128	-116
13	43	-30
14	37	-23
15	4	+11
16	168	-152
17	61	-44
18	492	-474
19	545	-526
20	118	-98
21	55	-34
22	13	+9
23	273	-250
24	22	+2
25	195	-170

Experiment

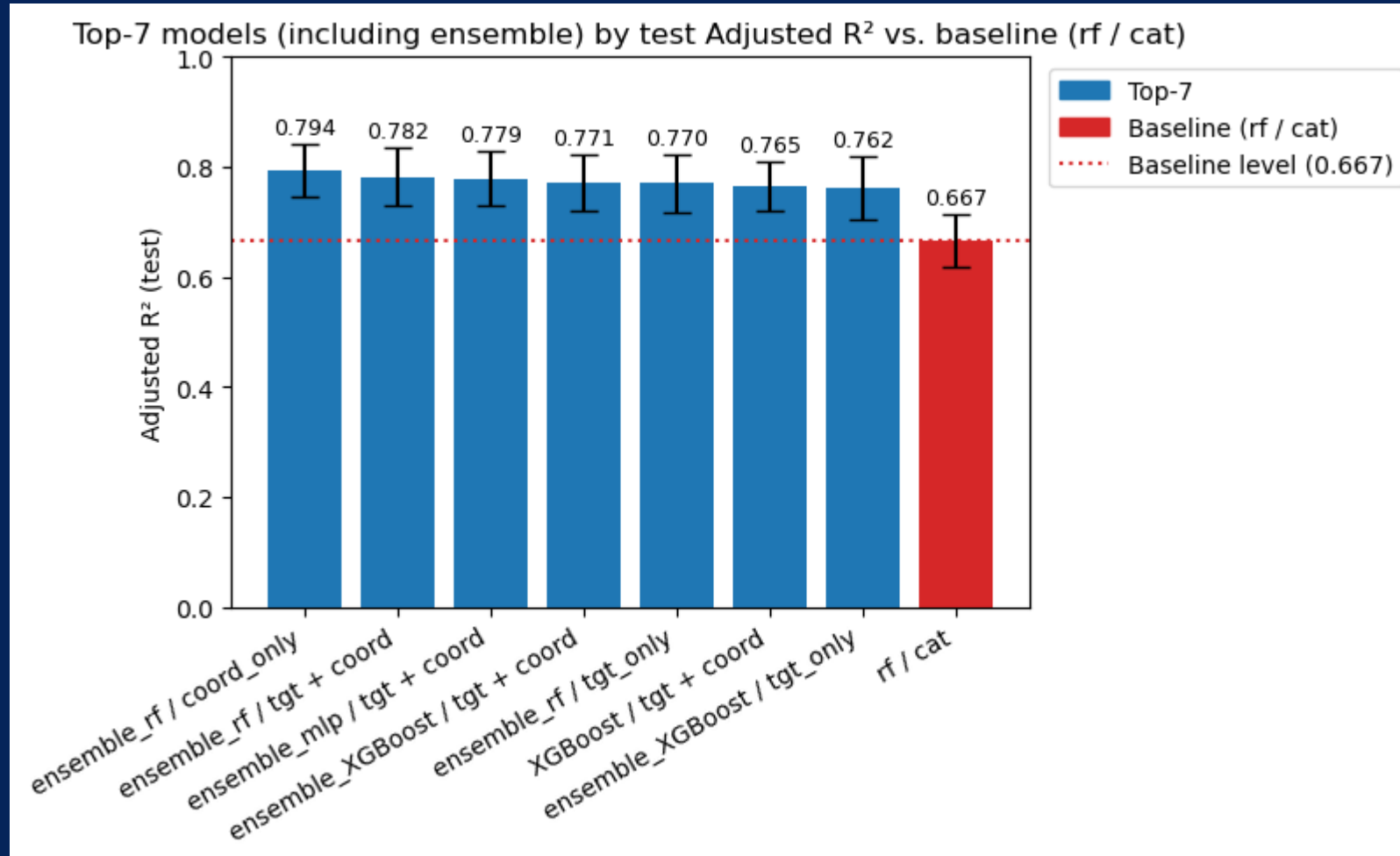
Proposal 2: Price-per-sqft

Idea: stacking with price-per-sqft

1. Train two model separately: $[M_p, M_{pp}]$
 - M_p predicts price directly
 - M_{pp} predicts $(\text{price}/\text{living_sqft}) = \text{price_per_sqft}$
2. Predict final price with weighted sum of model_price & model_ppsqft
 - $(\text{Final price}) = \lambda M_p(x) + (1 - \lambda) M_{pp}(x) * \text{living_sqft}$

Result

Result 2: stacking works especially well with RF



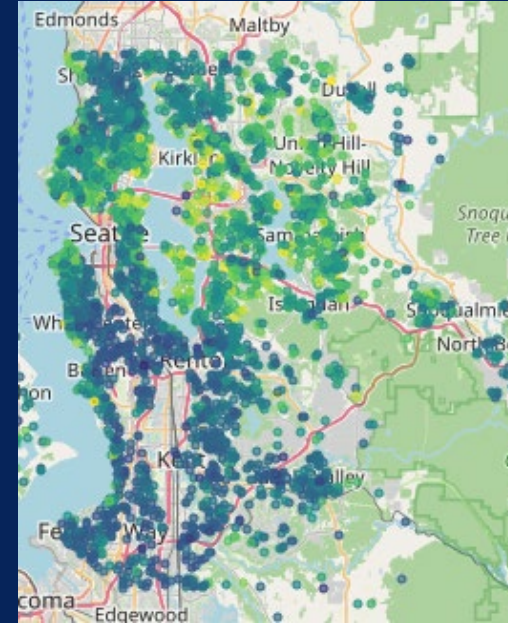
Conclusion

Conclusion:

1. Effectiveness of feature engineering methods depends on model type.
2. “Deeper relationship” can emerge after domain-aware preprocessing.

Limitations:

1. Only tested on one dataset
2. Lack of generalizability (Coordinate encoding & “per-sqft”)



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USA House Prices

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Conclusion

A white line graphic consisting of a horizontal segment followed by a diagonal segment extending towards the top right corner of the slide.

Thank you